Ahmed Salah Muhammed

GIS Developer | full-stack Developer | AWS Certified Cloud Practitioner
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PROFESSIONAL SUMMARY

GIS Developer and Urban Planner with a background in Environmental Planning from Cairo University, currently completing the ITI 9-month professional program in Geo-informatics. AWS Certified Cloud Practitioner experienced in spatial data analysis, ArcGIS Enterprise administration, cloud-based geospatial systems on AWS, Python automation, RESTful Map & Feature Services, 3D urban visualization, and full-stack GIS web development. Focused on integrating GIS with cloud computing, AI and automation, and smart-city solutions.

Focus areas: GIS Development · Cloud (AWS) · Smart Cities · Digital Twin · Spatial Analysis · Remote Sensing · AI Automation

EDUCATION

Diploma in IT — Geo-informatics — ITI (Intake 46) **10/2025 – 08/2026**
Information Technology Institute (ITI), 9-Month Professional Program · Smart Village, Cairo

B.Sc. in Urban & Regional Planning **10/2019 – 07/2024**
Cairo University — Faculty of Urban & Regional Planning, Dept. of Environmental Planning & Sustainable Infrastructure · Giza, Egypt

- GPA: 3.01 / 4.0 (B+) — ranked 5th in cohort · Graduation Project grade: A−

PROFESSIONAL EXPERIENCE

GIS Engineer — Etqaan Urban Planning Solutions **01/2025 – 10/2025**
Full-Time, On-site · Smart Village, Egypt

- Conducted spatial data analysis to support urban-planning and GIS studies.
- Designed and deployed Survey123 forms for field data-collection teams operating in Saudi Arabia.
- Administered ArcGIS Online (AGOL), Portal, ArcGIS Online Assistant, and related Esri products.
- Developed Python scripts to automate repetitive GIS workflows and improve operational efficiency.
- Produced 3D urban visualizations using CityEngine and Twinmotion.
- Prepared site sketches and planning layouts in collaboration with the Saudi Heritage Commission.

Urban Planner & GIS Analyst — Project Consultant Group **08/2023 – 09/2023**
Part-Time, Remote · Riyadh & Al-Nammas, Saudi Arabia

- Conducted a comprehensive flood-simulation study for Riyadh using spatial and hydrological analysis to assess flood-risk areas.
- Built and structured geodatabases for efficient spatial data management and analysis.
- Performed environmental and remote-sensing studies including topographic analysis, NDVI assessment, and terrain modeling.
- Contributed to urban GIS studies for Al-Nammas, supporting spatial planning and environmental evaluation.

KEY PROJECTS

Origins of Urban Heritage in Saudi Arabian Regions (KSA) **05/2025 – 10/2025**
Full-Time, On-site (Smart Village) · Etqaan Urban Planning Solutions · GIS Analyst & Urban Planner

- Led the geospatial data lifecycle for a nationwide heritage-documentation initiative — set spatial standards for site identification, managed ArcGIS Enterprise synchronization, designed field-survey logic, and ran advanced spatial queries to safeguard the national heritage database.

Old Town Buildings Registration — AlUla City (KSA) **01/2025 – 5/2025**
Full-Time, On-site (Smart Village) · Etqaan Urban Planning Solutions · GIS Analyst & Urban Planner

- Architected the geospatial backend for AlUla Old Town registration — published optimized RESTful Map & Feature Services, engineered a relational schema linking spatial features to AutoCAD files, PDFs, and multimedia, and built a dynamic workflow tracking construction progress via completion-based symbolization.

Geo-GenAI Hub Final Project, ITI-46-GeoGenAI-Hub **Graduation Project**

- A GIS platform that lets professionals interact with map data through natural language instead of traditional, complex GIS tooling — chat with your data, review and edit it safely, and query technical documentation grounded in your actual spatial layers.

ArcGIS Salah MCP, ArcGIS Salah MCP Add-in **Personal Project**

- Built an ArcGIS Pro Add-in integrating Claude AI with GIS via the Model Context Protocol (MCP) — a bridge between ArcGIS and Claude Desktop, enabling map operations, layer management, and spatial queries, making web apps also dashboards and deploying into GitHub through natural language.

Flood Simulation — Salloum, Egypt, Simulation Video **Personal / Academic**

- Modeled rainfall, flash floods, and inundation for Salloum using ArcGIS Pro, and produced a public YouTube tutorial covering the full pipeline from data preparation to 3D visualization.

NOSEJ — The Digital Boutique, React.js nosej **Personal Project**

- Designed and built a responsive online boutique presenting a curated collection of high-end essentials, deployed on Vercel.

PUBLICATIONS

Integrated Environmental Assessment of Sustainability Land Suitability for Agricultural Development in the New Delta Region Using a Bio-Economy Mapping Approach and Remote Sensing
Qutb, S., **Salah, A.**, & Taher, N. S. (2025). **Aswan University Journal of Environmental Studies (AUJES)**
A comprehensive environmental assessment of urban-development priorities within Egypt's New Delta Project using a bio-economy mapping approach, addressing food security, environmental sustainability, and spatial planning.

TECHNICAL SKILLS

GIS & Geospatial
ArcGIS Pro · ArcMap · ArcGIS Online (AGOL) · ArcGIS Enterprise & Portal · ArcGIS Enterprise Administration · ArcGIS Urban · QGIS · Post GIS · Survey123 · CityEngine · Feature & Map Services · Geodatabase Design · Spatial Analysis · Remote Sensing (NDVI, Terrain Modeling) · Flood Simulation · GeoAI · Google Earth Engine

Cloud & Infrastructure (AWS)
AWS (Certified Cloud Practitioner) · EC2 · EBS Multi-Attach · EFS · ArcGIS Enterprise on AWS theoretically · Cloud Architecture · Scalable Systems

Programming & Development
Python · Big Data and ML · ArcPy · C# · .NET / ASP.NET Core · JavaScript · HTML5 · CSS · Tailwind CSS · REST APIs · Swagger / OAS · Esri Maps SDK for JS · Esri Maps SDK for .Net · MERN stack (React.js – Node.js – MongoDB – Express.js), PostgreSQL, supabase

AI & Automation
AI Agents & Workflows · n8n · Model Context Protocol (MCP) · Claude AI Integration · Function Calling · Fine Tuning · Hugging Face · GenAi · Lang Chain

Foundations
Object-Oriented Programming (OOP) · DBMS · Networking · Operating Systems · Cybersecurity Fundamentals

CERTIFICATIONS & TRAINING

- **Cloud (AWS):**
  - **AWS Certified Cloud Practitioner (May 2026 – May 2029).**
  - AWS Academy Graduate — Cloud Foundations, CT University (Mar 2026).
- **Esri North Africa:** Water Utility Network; Image Analysis; GeoAI; ArcGIS Enterprise; ArcGIS Online.
- **Leadership:**
  - Aspire Leaders Program 2026 — Aspire Institute (Apr 2026).
  - McKinsey Forward Program — McKinsey & Company (Jun 2026).
- **GIS Development & APIs (Udemy):**
  - React.js with Johnas – Udemy | Feb 2026; MERN stack with Johnas – Udemy | May 2026; ArcPy for Python Developers using ArcGIS Pro (Apr 2026); APIs & RESTful APIs Crash Course (Apr 2026); Learn Swagger and the OAS (May 2026).
- **AI & Automation:**
  - AI Agents & Workflows — The Practical Guide, Udemy (May 2026); Complete n8n & AI Automation, Claude Code in Action, Anthropic (Apr 2026); Introduction to Claude Cowork, Anthropic (Apr 2026).

[My Certificates]

LANGUAGES

Arabic — Native · **English** — Professional working proficiency

ADDITIONAL INFORMATION

- **Extracurricular:** Football — registered player, Egyptian Football Association (former Fourth Division).
- **Military Service: Exempted**
- **Nationality:** Egyptian